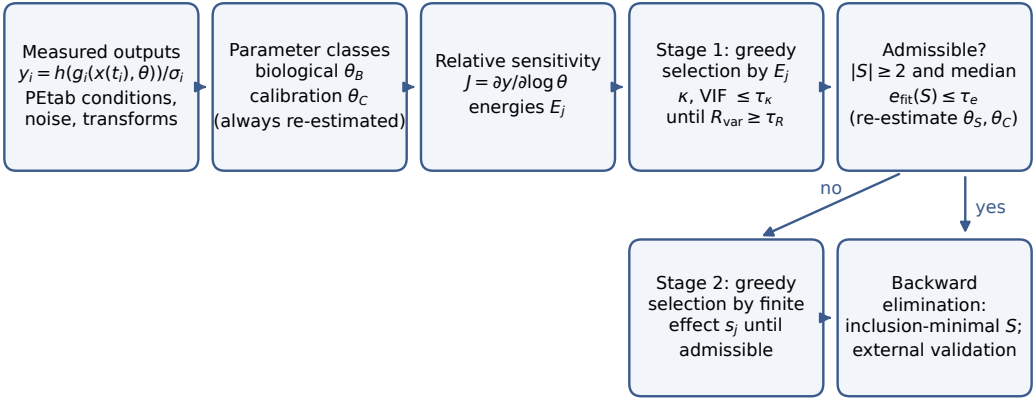
Measured outputs
$y_i = h(g_i(x(t_i), \theta))/\sigma_i$
PEtab conditions,
noise, transforms
Parameter classes
biological $\theta_B$
calibration $\theta_C$
(always re-estimated)
Relative sensitivity
$J = \partial y/\partial \log \theta$
energies $E_j$
Stage 1: greedy
selection by $E_j$
$\kappa$, VIF $\leq \tau_\kappa$
until $R_{\text{var}} \geq \tau_R$
Admissible?
$|S| \geq 2$ and median
$e_{\text{fit}}(S) \leq \tau_e$
(re-estimate $\theta_S, \theta_C$)
no
yes
Stage 2: greedy
selection by finite
effect $s_j$ until
admissible
Backward
elimination:
inclusion-minimal $S$;
external validation